correspond to those discovered by *Voyager* on Titan, and in the same proportions. The correspondence is one to one. The next most abundant gases that we've found in the laboratory will be looked for in future studies of Titan. The most complex organic gases we make have six or seven carbon and/or nitrogen atoms. These product molecules are on their way to forming tholins.

WE HAD HOPED FOR A BREAK in the weather as *Voyager 1* approached Titan. A long distance away, it appeared as a tiny disk; at closest approach, our camera's field of view was filled by a small province of Titan. If there had been a break in the haze and clouds, even only a few miles across, as we scanned the disk we would have seen something of its hidden surface. But there was no hint of a break. This world is socked in. No one on Earth knows what's on Titan's surface. And an observer there, looking up in ordinary visible light, would have no idea of the glories that await upon ascending through the haze and beholding Saturn and its magnificent rings.

From measurements by *Voyager*, by the *International Ultraviolet Explorer* observatory in Earth orbit, and by ground-based telescopes, we know a fair amount about the orange-brown haze particles that obscure the surface: which colors of light they like to absorb, which colors they pretty much let pass through them, how much they bend the light that does pass through them, and how big they are. (They're mostly the size of the particles in cigarette smoke.) The "optical properties" will depend, of course, on the composition of the haze particles.

In collaboration with Edward Arakawa of Oak Ridge National Laboratory in Tennessee, Khare and I have measured the optical properties of Titan tholin. It turns out to be a dead ringer for the real Titan haze. No other candidate material, mineral or organic, matches the optical constants of Titan. So we can fairly claim to have bottled the haze of Titan—formed high in its atmosphere, slowly falling out, and accumulating in copious amounts on its surface. What is this stuff made of?